

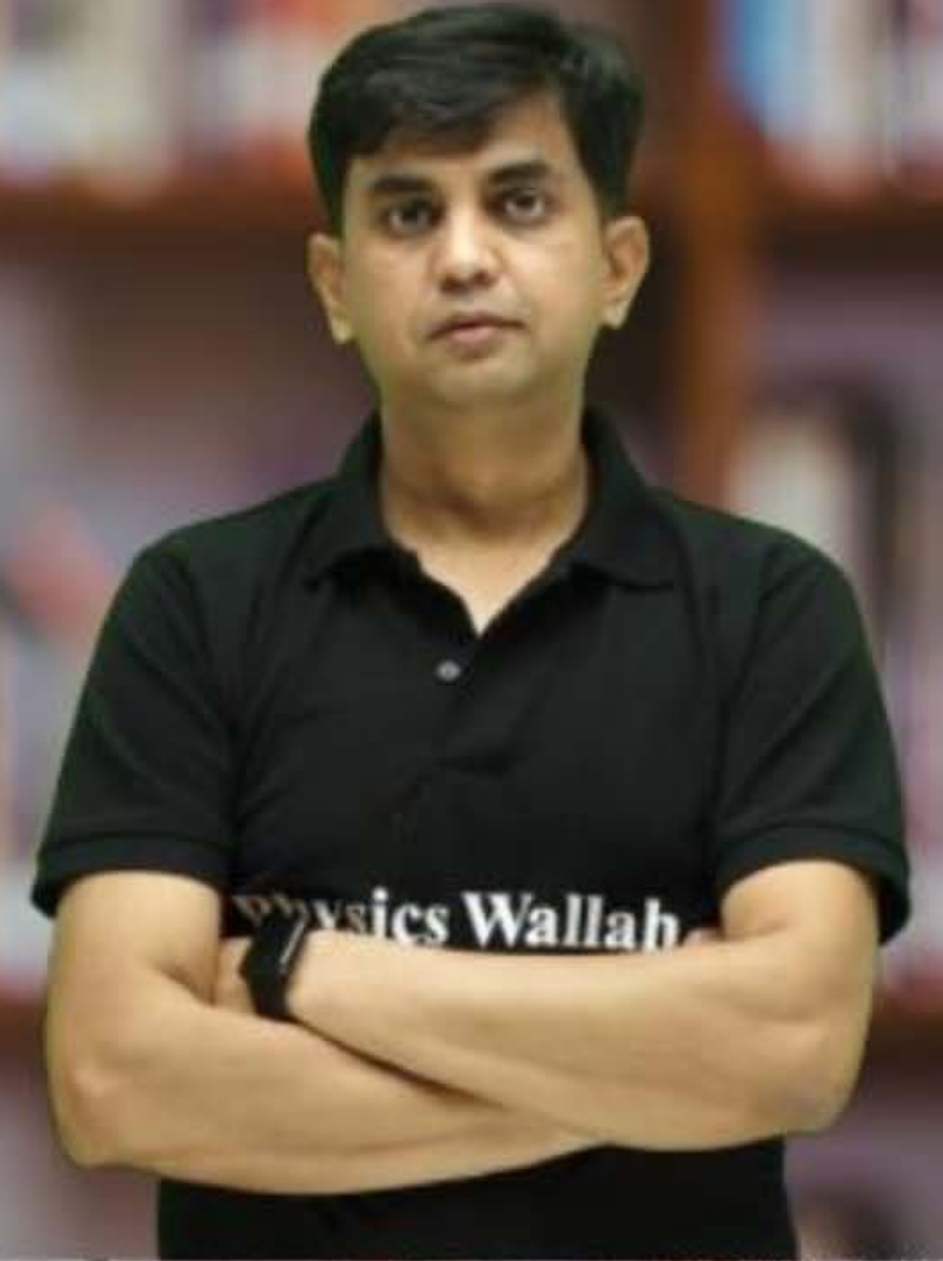
CS & IT ENGINEERING



Computer Network

Switching Performance Metrics

Lecture No. - 11



By - Abhishek Sir



Recap of Previous Lecture



Topic

Network Delays





Topics to be Covered



Topic

End-to-End Delays





Topic : End-to-End Delay



Transmitter
(Sender) ✓

Data Link Layer

1. Send Buffer
2. Receive Buffer

Physical Layer

Receiver ✓

Data Link Layer

1. Send Buffer
2. Receive Buffer

Physical Layer





Topic : End-to-End Delay ✓

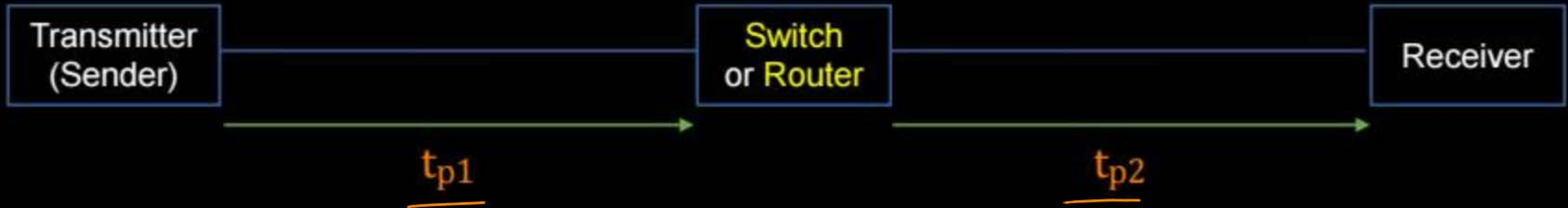


→ Time required for a packet to be transmitted from Transmitter to Receiver

→ One-way delay



Topic : Store-and-Forward Device



→ Suppose all links have uniform bandwidth



Topic : End-to-End Delay

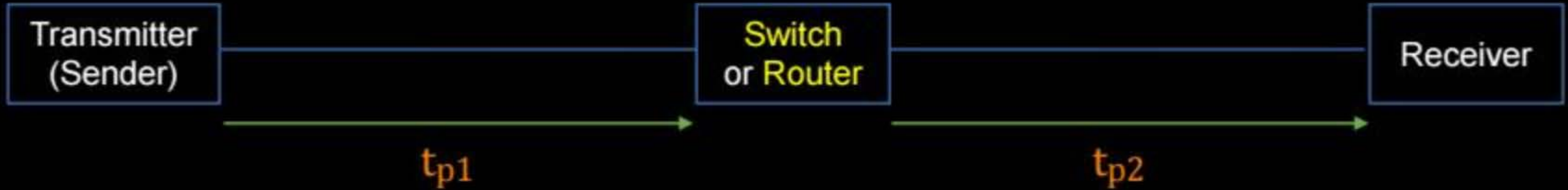
* For one packet

→ With one 'Store-and-Forward device'

$$\begin{aligned} \text{End-to-end delay} = & \text{Transmission delay} + \text{Propagation delay for first link} \\ & + \text{Store-and-Forward delay} \\ & + \text{Propagation delay for second link} \end{aligned}$$



Topic : Store-and-Forward Delay



$$\text{Store-and-Forward delay} = \text{Queuing delay at device} + \text{Processing delay by device} + \text{Transmission delay}$$



Topic : End-to-End Delay



→ Total time required for a file or message to be transmitted
from Transmitter to Receiver

$$\text{Number of packets (N)} = \frac{\text{File Size}}{\text{Payload Size}} = \frac{\text{File Size}}{\text{Packet Size}}$$

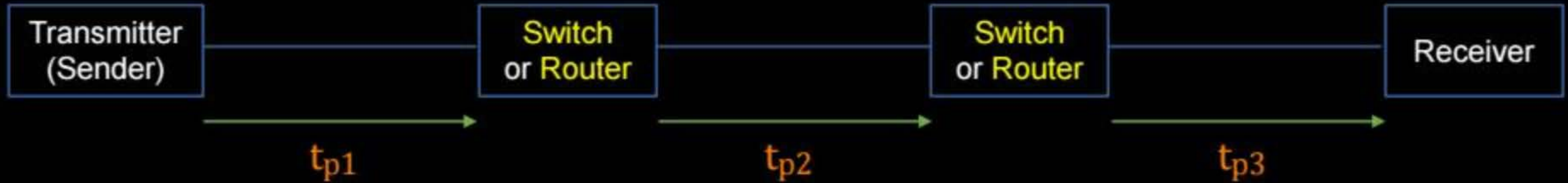
* Suppose sliding Window Protocol

$$\text{Transmit Window size} \geq N$$

* Suppose negligible size packet header



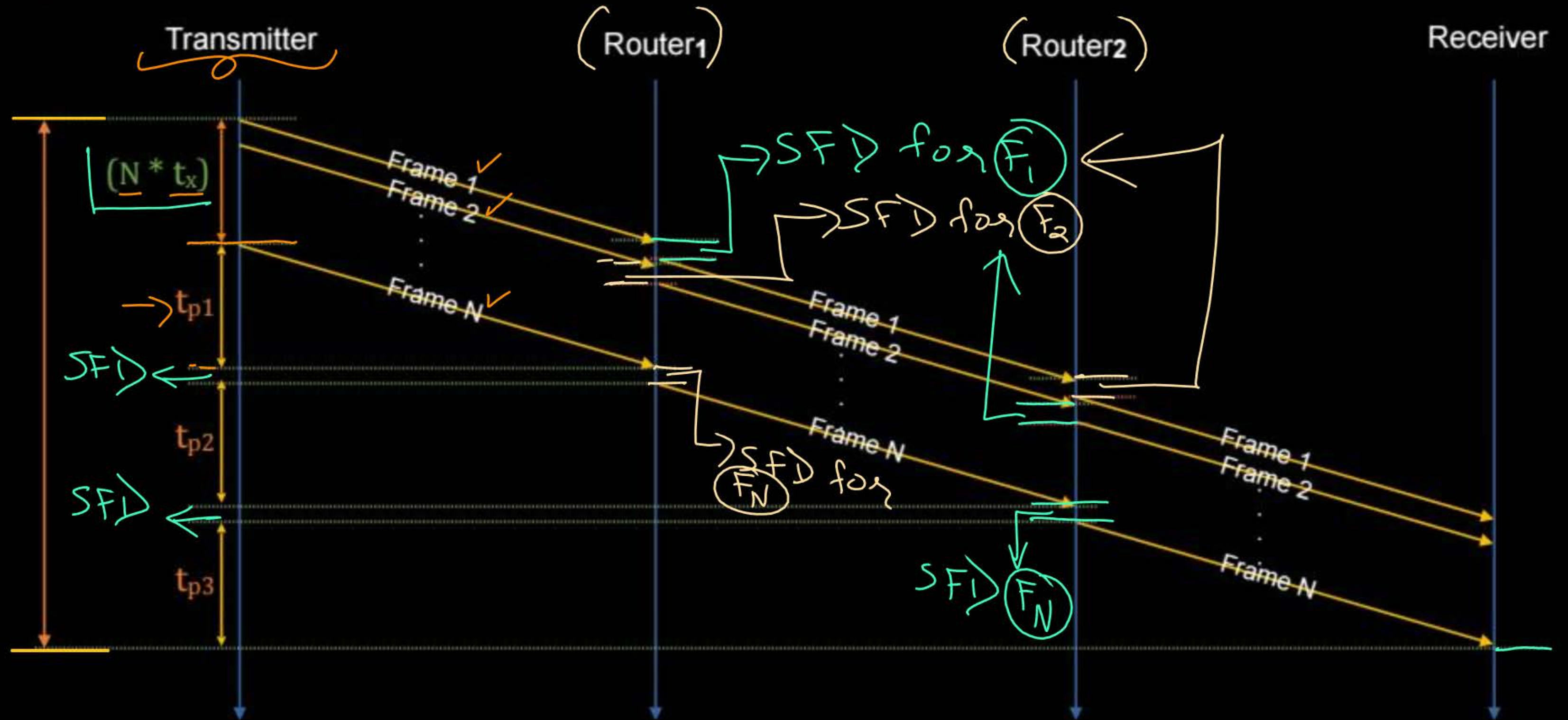
Topic : Store-and-Forward Device



→ Suppose all links have uniform bandwidth



Topic : End-to-End Delay





Topic : End-to-End Delay

→ With two 'Store-and-Forward device'

End-to-end delay = ($N * \text{Transmission delay}$) + Propagation delay for first link
+ Store-and-Forward delay
+ Propagation delay for second link
+ Store-and-Forward delay
+ Propagation delay for third link

$$= (N * t_x + t_{p_1}) + \text{SFD} + t_{p_2} + \text{SFD} + t_{p_3}$$



Topic : End-to-End Delay

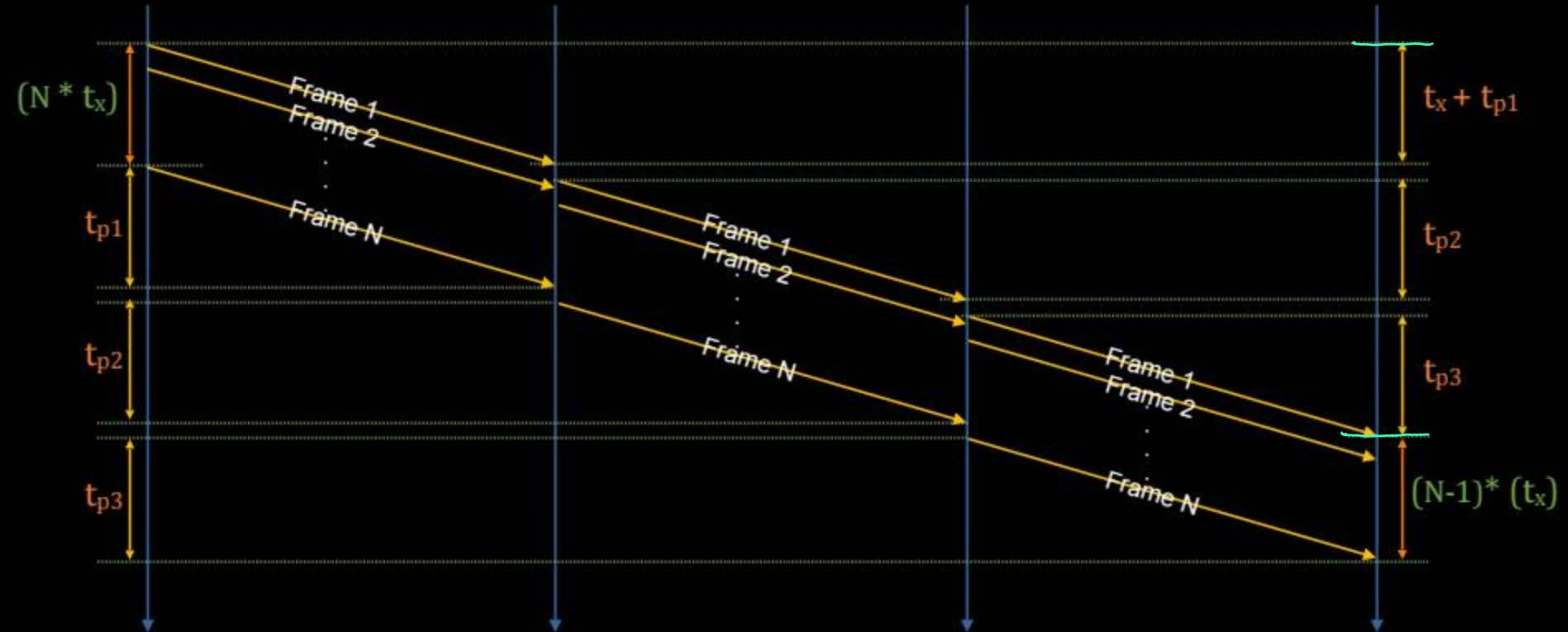


Transmitter

Router₁

Router₂

Receiver





Topic : End-to-End Delay



→ With two 'Store-and-Forward device'

$$\begin{aligned} \text{End-to-end delay} = & \underbrace{\text{Transmission delay}} + \text{Propagation delay for first link} \\ & + \text{Store-and-Forward delay} \\ & + \text{Propagation delay for second link} \\ & + \text{Store-and-Forward delay} \\ & + \text{Propagation delay for third link} \\ & + \underbrace{((N-1) * \text{Transmission delay})} \end{aligned}$$



Topic : End-to-End Delay



→ For one 'Store-and-Forward device'

$$\text{Store-and-Forward delay} = \text{Queuing delay at device} + \text{Processing delay by device} + \text{Transmission delay}$$

[MCQ] [GATE-2012] [2-Mark] IIT-D



#Q. Consider a source computer (S) transmitting a file of size 10^6 bits to a destination computer (D) over a network of two routers (R1 and R2) and three links (L1, L2, and L3). L1 connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D. Let each link be of length 100 km. Assume signals travel over each link at a speed of 10^8 meters per second. Assume that the link bandwidth on each link is 1 Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D.

Ans: A



☒ A 1005 ms

☐ B 1010 ms

☐ C 3000 ms

☐ D 3003 ms

Solution:-

$$\text{Packet Size} = 1000 \text{ bits} = (10^3 \text{ bits})$$

$$\text{Bandwidth} = 1 \text{ Mbps} = 10^6 \text{ bits / sec}$$

$$\begin{aligned} \text{Transmission Delay } [t_x] &= \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 1 \text{ ms} \\ &= 10^{-3} \text{ sec} \end{aligned}$$

$$\begin{aligned} \text{Distance} &= 100 \text{ Km} = (10^5 \text{ m}) \\ \text{Signal Speed} &= 10^8 \text{ m/s} \end{aligned}$$

$$\text{Propagation Delay } [t_{p1}] = \frac{\text{Distance}}{\text{Signal Speed}} = \frac{10^5 \cancel{\text{m}}}{10^8 \cancel{\text{m/s}}} = 1 \text{ ms}$$

$$= 10^{-3} \text{ sec}$$

$$t_{p1} = t_{p2} = t_{p3} = 1 \text{ ms}$$

$$\text{File Size} = 10^6 \text{ bits}$$

$$\text{Packet Size} = 10^3 \text{ bits}$$

$$\text{Number of packets (N)} = \frac{\text{File Size}}{\text{Packet Size}} = \frac{10^6 \text{ bits}}{10^3 \text{ bits}} = 10^3$$

$$\text{End-to-end delay} = (N * \text{Transmission delay}) + \text{Propagation delay for first link} \\ + \text{Store-and-Forward delay} \\ + \text{Propagation delay for second link} \\ + \text{Store-and-Forward delay} \\ + \text{Propagation delay for third link}$$

Q V.V. Imp.

CS-12

CS-14 **

CS-15

CS-24

 CS-05

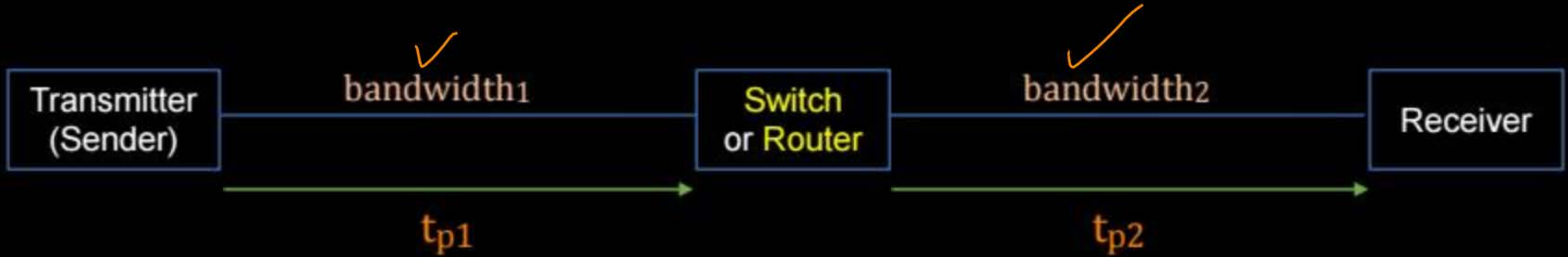
$$\begin{aligned}
 \text{End-to-end delay} = & \underbrace{(N * \text{Transmission delay})}_{\text{SFD}_1} + \underbrace{\text{Propagation delay for first link}}_{\text{SFD}_2} \\
 & + \underbrace{\text{Transmission delay}}_{\text{SFD}_2} + \underbrace{\text{Propagation delay for second link}}_{\text{SFD}_3} \\
 & + \underbrace{\text{Transmission delay}}_{\text{SFD}_3} + \underbrace{\text{Propagation delay for third link}}_{\text{SFD}_4}
 \end{aligned}$$

$$\text{End-to-end delay} = (N * t_x + t_{p1}) + \underbrace{(t_x + t_{p2})}_{\text{SFD}_1} + \underbrace{(t_x + t_{p3})}_{\text{SFD}_2}$$

$$\begin{aligned}
 &= (10^3 * 1 + 1) + (1 + 1) + (1 + 1) \text{ ms} \\
 &= 1005 \text{ ms}
 \end{aligned}$$



Topic : Store-and-Forward Device



↓ non-uniform
→ Suppose links have different bandwidth

Example :-

[NAT]



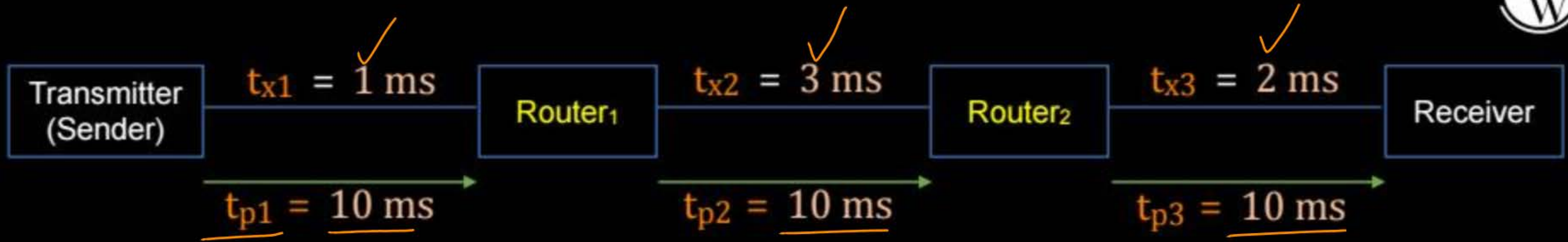
#Q. Consider two hosts A and B are connected through two routers along the path. The router takes negligible packet processing overhead and queue at router is sufficiently large. The packet transmission delay for first, second, and third link are 1, 3, and 2 milliseconds respectively. Each link propagation delay is 10 milliseconds. Host A wants to send a file of size 4000 bits to host B, by dividing the file into 1000 bits packets. Calculate the amount of time required (in milliseconds) to deliver the file completely from A to B?

$$\text{File Size} = 4000 \text{ bits}$$

$$\text{Packet Size} = 1000 \text{ bits}$$

$$\text{Number of packets (N)} = \frac{\text{File Size}}{\text{Packet Size}} = \frac{4000 \text{ bits}}{1000 \text{ bits}}$$

$$N = 4$$



<u>Packets</u> [Transmission start at $t = 0 \text{ ms}$]	Transmission completed by transmitter	Packet completely received by router1	Transmission completed by router1	Packet completely received by router2	Transmission completed by router2	Packet completely received by receiver
First Packet	1 ms	11 ms	11 - 14 ms	24 ms	24 - 26 ms	36 ms
Second Packet	2 ms	12 ms	14 - 17 ms	27 ms	27 - 29 ms	39 ms
Third Packet	3 ms	13 ms	17 - 20 ms	30 ms	30 - 32 ms	42 ms
Fourth Packet	4 ms	14 ms	20 - 23 ms	33 ms	33 - 35 ms	45 ms

$$N=4, \quad t_{x_1}=1\text{ms} \quad t_{x_2}=3\text{ms} \quad t_{x_3}=2\text{ms}$$
$$t_{p_1}=t_{p_2}=t_{p_3}=10\text{ms}$$



$$\text{End-to-End Delay} = (N * t_{x_2}) + t_{x_1} + t_{x_3} + (t_{p_1} + t_{p_2} + t_{p_3})$$

$$= (4 * 3\text{ms}) + 1\text{ms} + 2\text{ms} + (10 + 10 + 10)\text{ms}$$
$$= 45\text{ms}$$



Topic : Bottleneck Bandwidth



- Smallest link bandwidth along end-to-end network path
- Affect end-to-end network throughput



Topic : Bottleneck Bandwidth

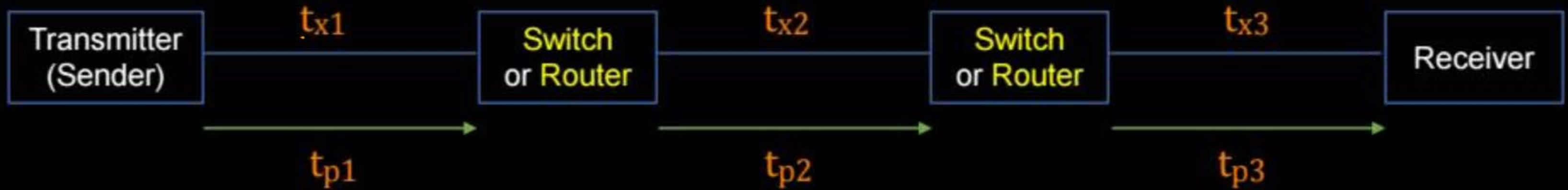
$$Tx \propto \frac{1}{\text{Bandwidth}}$$



$$\text{Bottleneck Bandwidth} = \min[\text{bandwidth}_1, \text{bandwidth}_2, \text{bandwidth}_3]$$



Topic : Store-and-Forward Device



→ Suppose links have different bandwidth



Topic : End-to-End Delay



→ Consider negligible processing and propagation delays

$$t_{P_1} = t_{P_2} = t_{P_3} = 0$$

→ Consider very small size packets
[Packet size is not given] → ✖✖

→ Assume $\text{bandwidth}_2 < \text{bandwidth}_3 < \text{bandwidth}_1$

$$t_{x_2} > t_{x_3} > t_{x_1}$$



Topic : End-to-End Delay



→ The time required to deliver very large size file from transmitter to receiver

N is very large

End-to-end delay

$$= (N * t_{x2}) + t_{x1} + t_{x3} + (t_{p1} + t_{p2} + t_{p3})$$

$$= (N * t_{x2}) + t_{x1} + t_{x3}$$

$$= (N * t_{x2})$$

$$(N * t_{x2}) \gg \underbrace{(t_{x1} + t_{x3})}_{\text{Negligible}}$$



Topic : End-to-End Delay

V.V. Imp.



$$\begin{aligned}\text{End-to-end delay} &= (N * t_{x2}) = \frac{\text{File size}}{\text{Packet size}} * \frac{\text{Packet size}}{\text{Bandwidth}_2} \\ &= \frac{\text{File size}}{\text{Bandwidth}_2}\end{aligned}$$

$$\text{End-to-End Delay} = \frac{\text{File size}}{\text{Bottleneck Bandwidth}}$$

[MCQ] [GATE-2026, Set-2] [1-Mark]



#Q. Consider a file of size 4 million bytes being transferred between two hosts connected via a path consisting of three consecutive links of bandwidth 2 Mbps, 500 kbps, and 1 Mbps, respectively. All processing delays and propagation delays are negligible. Assume that there is no other background traffic over the path and no other additional overhead to transfer the file. Which one of the following is the total time (in seconds) to transfer the file?
Note: $1\text{M}=10^6$, $1\text{k}=10^3$

- ☐ A 8
- ☐ B 16
- ☒ C 64
- ☐ D 731

$$\begin{aligned}\text{File Size} &= 4 \text{ million bytes} \\ &= 4 \times 10^6 \text{ bytes}\end{aligned}$$

$$\begin{aligned}\text{Bottleneck Bandwidth} &= \min[2 \text{ Mbps}, 500 \text{ kbps}, 1 \text{ Mbps}] \\ &= 500 \text{ kbps}\end{aligned}$$

End-to-End Delay

$$= \frac{\text{File Size}}{\text{Bottleneck Bandwidth}}$$

$$\begin{aligned}&= \frac{4 \times 10^6 \text{ bytes}}{500 \text{ kbps}} = \frac{32 \times 10^6 \text{ bits}}{500 \times 10^3 \text{ bits/sec}} \\ &= 64 \text{ sec}\end{aligned}$$

Ans: C



2 mins Summary



Topic

End-to-End Delays

Performance:-

* Delay

* Throughput

* Reliability



THANK - YOU